

Telecom Customer Churn Analysis - Project Documentation Summary

Project Overview:

Customer churn is one of the most critical business challenges in the telecommunications industry, as acquiring new customers is significantly more expensive than retaining existing ones. The objective of this project was to analyze customer demographics, subscription patterns, service usage, contract information, and payment preferences to identify the key drivers of customer churn and uncover actionable insights for improving customer retention.

The analysis was performed using Python-based data analytics libraries, including Pandas, NumPy, Matplotlib, and Seaborn. Through data cleaning, exploratory data analysis (EDA), and visualization techniques, the project examined relationships between various customer attributes and churn behavior.

Dataset Preparation and Data Cleaning:

Before performing the analysis, the dataset underwent several preprocessing steps to ensure data quality and consistency:

- Imported and inspected the telecom customer dataset.
- Replaced blank values in the **TotalCharges** column with 0 for customers whose tenure was 0 months.
- Verified and handled data type inconsistencies.
- Checked for missing values and duplicate records.
- Converted the **SeniorCitizen** field into a more interpretable categorical format for analysis.

These steps ensured that the dataset was accurate and suitable for exploratory analysis.

Exploratory Data Analysis (EDA)

Overall Churn Distribution

The initial analysis focused on understanding the overall churn rate within the customer base.

Key Finding:

- Approximately **26.54% of customers had churned**, while **73.46% remained active**.

This indicates that although the majority of customers are retained, a significant proportion discontinue services, making churn reduction an important business objective.

Demographic Analysis

Gender and Churn

Customer churn was analyzed across gender categories.

Observation:

- Churn rates were relatively similar across male and female customers.
- Gender did not appear to be a significant factor influencing customer retention.

Business Insight:

- Retention strategies should focus more on behavioral and service-related factors rather than gender-based segmentation.

Senior Citizens and Churn

The relationship between customer age category (Senior Citizen status) and churn was examined.

Observation:

- Senior citizens exhibited a noticeably higher churn rate than non-senior customers.

Business Insight:

- Senior customers may face challenges related to service complexity, pricing, or support accessibility.
- Targeted retention programs and customer support initiatives could help improve retention within this segment.

Customer Tenure Analysis

Tenure represents the duration for which a customer has remained subscribed to the company's services.

Observation:

- Customers with very short tenures (1–2 months) showed significantly higher churn.
- Long-tenured customers demonstrated stronger loyalty and were less likely to churn.

Business Insight:

- The early stages of the customer lifecycle represent the highest-risk period.
- Enhanced onboarding programs and proactive engagement during the first few months can help improve retention.

Contract Type Analysis

Customer churn was analyzed across different contract categories.

Observation:

- Customers on **Month-to-Month contracts** experienced the highest churn rates.
- Customers with **One-Year** and **Two-Year contracts** were substantially more likely to remain with the company.

Business Insight:

- Long-term contracts contribute significantly to customer retention.
- Businesses should encourage customers to migrate from month-to-month plans to longer contract periods through incentives and loyalty benefits.

Service Usage Analysis

Several telecom service offerings were analyzed to determine their impact on customer churn.

Internet Service

Observation:

- Customers using **Fiber Optic Internet** showed the highest churn rates.
- Customers without internet service exhibited the lowest churn.

Business Insight:

- Further investigation into pricing, service quality, or customer expectations for Fiber Optic services may be necessary.
- Fiber Optic customers represent a high-risk segment requiring focused retention efforts.

Value-Added Services

The following services were analyzed:

- Online Security
- Online Backup
- Device Protection
- Tech Support

Observation:

- Customers who did not subscribe to these services experienced significantly higher churn rates.
- Customers enrolled in these services demonstrated stronger retention.

Business Insight:

- Value - added services increase customer engagement and create switching barriers.
- Promoting bundled service packages could reduce churn and improve customer lifetime value.

Streaming Services

The impact of:

- Streaming TV
- Streaming Movies

was also examined.

Observation:

- Churn differences between subscribers and non-subscribers were relatively small.

Business Insight:

- Streaming services alone do not appear to strongly influence retention decisions.
- They may contribute to customer satisfaction but are not major churn drivers.

Phone Services and Multiple Lines

Observation:

- Customers with or without phone services displayed similar churn patterns.
- Multiple lines showed limited impact on churn behavior.

Business Insight:

- These services are not primary predictors of customer attrition.

Payment Method Analysis

Customer churn was evaluated across different payment methods.

Observation:

- Customers paying through **Electronic Check** demonstrated the highest churn rates.
- Other payment methods showed comparatively lower churn.

Business Insight:

- Electronic Check users may represent a higher-risk customer segment.
- Encouraging automatic payment options and digital billing solutions may improve retention.

Key Business Drivers of Churn

The analysis identified the following factors as the strongest contributors to customer churn:

1. Month-to-Month Contracts
2. Short Customer Tenure
3. Fiber Optic Internet Service
4. Lack of Online Security
5. Lack of Online Backup
6. Lack of Device Protection
7. Lack of Tech Support
8. Electronic Check Payment Method
9. Senior Citizen Customer Segment

Conclusion

The analysis revealed that customer churn is driven primarily by contract flexibility, tenure, service adoption, and payment preferences rather than demographic factors such as gender. Customers on month-to-month contracts, those with short tenures, Fiber Optic users, and

customers who do not subscribe to support-related services exhibit the highest likelihood of churn.

To reduce churn and improve customer retention, the company should focus on promoting long-term contracts, increasing adoption of value-added services, enhancing support for senior citizens, improving onboarding experiences for new customers, and implementing targeted retention campaigns for high-risk customer segments. These initiatives can help improve customer loyalty, reduce revenue loss, and increase overall customer lifetime value.